

PP-narrow operators on L_1 : the sum problem is false

Literature-already-answered note for arXiv:math/0311324

12 August 2026

Status. The general-range sum question has a negative answer. Mykhaylyuk–Popov constructed a Banach space Y and PP-narrow operators $T_1, T_2 : L_1[0, 1] \rightarrow Y$ for which $T_1 + T_2$ is not PP-narrow. Their result appeared in *J. Math. Anal. Appl.* 404 (2013), 554–561, doi:10.1016/j.jmaa.2013.03.008. This packet records that later-literature counterexample and makes no novelty claim.

The source question

Let (Ω, Σ, μ) be a nonatomic probability space. A *sign* on $A \in \Sigma$ is

$$f = \mathbf{1}_{B_1} - \mathbf{1}_{B_2}, \quad A = B_1 \dot{\cup} B_2, \quad \mu(B_1) = \mu(B_2).$$

An operator $T : L_1(\Omega) \rightarrow X$ is PP-narrow when, for every positive-measure A and every $\varepsilon > 0$, some sign f on A satisfies $\|Tf\| < \varepsilon$.

The source paper notes on PDF page 4 that PP-narrow operators on L_p are not closed under sums when $1 < p < \infty$, and asks whether closure under sums might nevertheless hold for $p = 1$. The surrounding setup keeps the range X arbitrary. Thus the selected question is

For every Banach space X , must two PP-narrow maps $L_1[0, 1] \rightarrow X$ have PP-narrow sum?

Let h_{-1} and h_1 be signs supported on A_{-1} and A_1 respectively with $\|Th_{\pm 1}\| \leq \varepsilon_{\pm 1}$; put

$$\begin{aligned} A_{-1,-1} &= \{h_{-1} = -1\}, & A_{-1,1} &= \{h_{-1} = 1\}, \\ A_{1,-1} &= \{h_1 = -1\}, & A_{1,1} &= \{h_1 = 1\} \end{aligned}$$

and continue in the above fashion. This yields part (a).

Part (b) follows from (a) and Remark 2.2(b). □

For $1 < p < \infty$ the class of PP-narrow operators on L_p is not stable under taking sums (see [8], p. 59); this is why we have to consider a smaller class of operators that we introduce next. Incidentally, the stability of PP-narrow operators on L_1 under sums is still an open problem.

Definition 2.4. An operator $T : L_p \rightarrow X$ is said to be *hereditarily PP-narrow* (HPP-narrow for short) if for every $A \in \Sigma^+$ and every nonatomic sub- σ -algebra $\Sigma_1 \subset \Sigma$ on A the restriction of T to $L_p(A, \Sigma_1, \mu)$ is PP-narrow.

Since every compact operator on L_p is PP-narrow and compactness is inherited by restrictions, compact operators on L_p are HPP-narrow. On the other hand, the operator

Source evidence: arXiv:math/0311324, PDF page 4 (unaltered crop).

The later counterexample

Mykhaylyuk–Popov prove the stronger assertion

for every Köthe Banach space E on $[0, 1]$, there are a Banach space Y and narrow operators $T_1, T_2 : E \rightarrow Y$ such that $T_1 + T_2$ is not narrow. (MP)

Their publisher abstract explicitly says that this answers the $E = L_1$ question in the negative. Before publication, the same authors stated the exact arbitrary-range problem in their 2012 Stefan Banach conference abstract and announced negative answers.

operators has a nice description in a more general setting of narrow operators on vector lattices. In [2] O. Maslyuchenko and the authors extended the notion of narrow operators to vector lattices and proved that, given order continuous Banach lattices E, F with E atomless, the set $N_r(E, F)$ of all regular narrow operators from E to F is a band in the vector lattice $L_r(E, F)$ of all regular operators from E to F . In particular, a sum of two regular narrow operators from E to F is narrow. Since all linear continuous operators on L_1 are regular, this covers the result on narrowness of a sum of two narrow operators on L_1 . On the other hand, every example of a non-narrow sum of two narrow operators on a rearrangement invariant space on $[0, 1]$ with an unconditional basis involves non-regular summands. However, the following problem has been remained unsolved [1].

Problem 1. *Let X be any Banach space. Is the sum of two narrow operators in $\mathcal{L}(L_1, X)$ narrow?*

Another problem concerns the classical atomless Banach lattice L_∞ . Since L_∞ is not order continuous, Theorem 1 cannot be applied to operators defined on L_∞ . Moreover, it was shown in [2] that the set of all narrow regular operators on L_∞ is not a band in $L_r(L_\infty)$. Nevertheless, the following problem remained open [2].

Problem 2. *Is a sum of two regular narrow operators on L_∞ narrow?*

We give negative answers to these problems.

Primary evidence: Mykhaylyuk–Popov conference abstract, PDF page 27 (unaltered crop).

Formal deduction from (MP)

Take $E = L_1[0, 1]$ in (MP). The term “narrow” used there is the Plichko–Popov notion called “PP-narrow” by the source paper: both definitions quantify over every measurable A , every $\varepsilon > 0$, and equal-measure signs on A . Hence the theorem supplies a Banach space Y and two PP-narrow maps $T_1, T_2 : L_1[0, 1] \rightarrow Y$.

The assertion that $T_1 + T_2$ is not narrow is precisely the negation of the source property. Equivalently, there are a positive-measure set A and $\varepsilon_0 > 0$ such that

$$\|(T_1 + T_2)f\| \geq \varepsilon_0 \quad \text{for every equal-measure sign } f \text{ on } A.$$

Thus T_1, T_2 answer the displayed source question negatively. Only two summands are needed.

Why there is no contradiction with the $L_1 \rightarrow L_1$ theorem

The range is decisive. For self-maps, the class of PP-narrow operators in $\mathcal{L}(L_1)$ is stable even under pointwise unconditional sums. The counterexample instead permits a specially chosen Banach range Y .

The 2025 monograph by Kadets–Martín–Rueda Zoca–Werner states both facts side by side and identifies the counterexample as Mykhaylyuk–Popov [236]:

operator on a space with the Daugavet property, to be discussed in Chapter 6. Let us introduce the necessary notation and definitions.

In this section (Ω, Σ, μ) is a finite or σ -finite nonatomic measure space. We denote by Σ^+ the collection of those $A \in \Sigma$ for which $0 < \mu(A) < \infty$. For $1 \leq p < \infty$, L_p means $L_p(\Omega, \Sigma, \mu)$, and $L_p^0(\Omega, \Sigma, \mu) \subset L_p$ is the subspace of all $f \in L_p$ with $\int_\Omega f d\mu = 0$.

DEFINITION 5.4.2. Let $A \in \Sigma^+$. A measurable function $f: \Omega \rightarrow \mathbb{R}$ is said to be a *sign on A* if it is of the form $f = \mathbb{1}_{B_1} - \mathbb{1}_{B_2}$, where B_1 and B_2 form a partition of A into two measurable subsets with $\mu(B_1) = \mu(B_2)$.

An operator $T \in L(L_p, X)$ is said to be *PP-narrow* if for every $A \in \Sigma^+$ and $\varepsilon > 0$ there is a sign f on A such that $\|Tf\| \leq \varepsilon$.

For the space L_1 the complement of this class of operators was studied under the name “sign-preserving operators” in papers by Ghousoub and Rosenthal [118], [262] and [263].

From the definition it follows that a PP-narrow operator cannot be bounded below. Also, every compact operator $T \in L(L_p, X)$, $1 \leq p < \infty$, is PP-narrow [252]. Indeed, for $A \in \Sigma^+$ consider a Rademacher sequence (r_n) on A , that is, a sequence of independent random variables on the probability space $(A, \Sigma|_A, \frac{\mu}{\mu(A)})$ with $\mu(\{r_n = 1\}) = \mu(\{r_n = -1\}) = \mu(A)/2$. (Cf. page 119.) All r_n are signs on A , $r_n \rightarrow 0$ weakly in L_p , so $\|Tr_n\| \rightarrow 0$ by compactness of T .

It is known [176, 286] that the class of PP-narrow operators in $L(L_1)$, is stable under the operation *unc*. Unfortunately, this result does not extend to those PP-narrow operators that act from L_1 into another space. Moreover, Mykhaylyuk and Popov [236] constructed a space Y and a couple of PP-narrow operators $T_1, T_2 \in L(L_1[0, 1], Y)$ whose sum is not PP-narrow; so in general it does not make sense to speak about infinite sums of PP-narrow operators. In order to solve this problem, we introduce a subclass of PP-narrow of operators, called hereditarily PP-narrow (HPP-narrow for short).

Status evidence: *Banach spaces with the Daugavet property*, Definition 5.4.2 and PDF page 158/book page 140 (unaltered crop).

Setting	Status	Reason
$T_1, T_2 : L_1 \rightarrow L_1$	positive	the self-map class is stable under <i>unc</i>
$T_1, T_2 : L_1 \rightarrow Y$, arbitrary Y	negative	Mykhaylyuk–Popov choose a range Y with a non-narrow sum
HPP-narrow maps $L_1 \rightarrow Y$	positive	hereditary narrowness is stable under sums and unconditional series

Proof intuition

PP-narrowness is an existential small-sign condition. Given T_1 and T_2 separately, the sign that makes T_1 small need not be the sign that makes T_2 small. A target norm can preserve this incompatibility: each coordinate test can be defeated on its own, while the combined operator remains uniformly large on all signs of one measurable set. That is the mechanism made possible by the custom range Y in the 2013 construction.

The source paper's repair is hereditary PP-narrowness. It demands the same small-sign property after restriction to every nonatomic sub- σ -algebra. One may first choose a sub- σ -algebra on which one summand is uniformly small, then use hereditary narrowness of the other inside that same algebra. This synchronizes the sign choices and yields

$$U \text{ PP-narrow, } V \text{ HPP-narrow} \implies U + V \text{ PP-narrow.}$$

The later counterexample shows that the hereditary hypothesis is substantive, not a technical convenience.

Scope and audit trail

The 2013 theorem is stronger than needed because it works for every Köthe Banach domain E ; the source question follows by the single specialization $E = L_1$. No claim is made here to reproduce the internal construction of Y . The literature implication and the matching of definitions and quantifiers are complete.

Human review. Check the source question on source PDF page 4; the exact formulation and negative announcement on conference PDF page 27; and the finished counterexample statement on monograph PDF page 158 (book page 140). The publisher record for the 2013 article should be checked under the DOI below.

References

- [1] V. Kadets, N. Kalton, and D. Werner, *Unconditionally convergent series of operators and narrow operators on L_1* , Bull. Lond. Math. Soc. 37 (2005), 265–274; arXiv:math/0311324.
- [2] V. V. Mykhaylyuk and M. M. Popov, *On sums of narrow operators on Köthe function spaces*, J. Math. Anal. Appl. 404 (2013), 554–561, doi:10.1016/j.jmaa.2013.03.008.
- [3] V. V. Mykhaylyuk and M. M. Popov, *On sums of narrow operators on Köthe Banach spaces*, abstract in the International Conference dedicated to the 120th anniversary of Stefan Banach, L'viv, 2012, p. 27.
- [4] V. Kadets, M. Martín, A. Rueda Zoca, and D. Werner, *Banach spaces with the Daugavet property*, preliminary version, 2025, Section 5.4; the counterexample paper is reference [236].